

PrimeNet Twin Prime Census: 1-3T

Formal Census and Validation Report

5,173,760,785 twin-prime events were observed among **108,340,298,703** verified prime-indexed outgoing gaps.

Global twin-event density: **0.047754721437340**

Instrument	PrimeNet Twin Prime Census v1.0.0
Event definition	$g(i) = 2$
Repository status	ACCEPTED
Validation status	PASSED
Numeric domain	1 to 3,000,000,000,000
Observation runtime	47.833 minutes

1. Provenance and repository acceptance

The census was performed only after the PrimeNet gap repository passed a full element-by-element acceptance audit. All 300 prime partitions and 300 gap partitions passed; 108,340,298,403 local gaps, 299 ordinary boundaries, and the single terminal boundary were verified with zero warnings and zero errors. The accepted contract is $g(i) = p(i+1) - p(i)$ for every stored prime index.

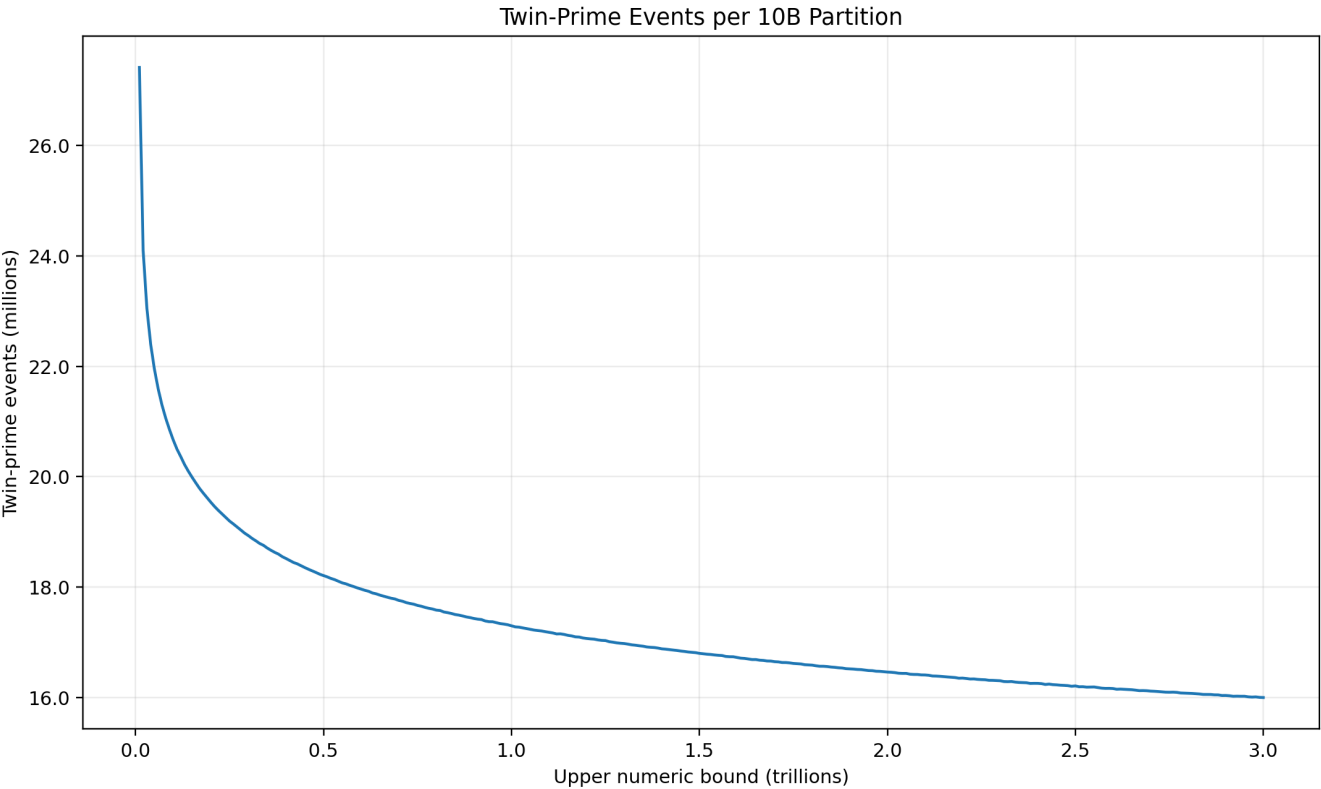
2. Independent census validation

Check	Status	Observed
partition_count	PASS	300
partition_sequence	PASS	1..300
numeric_range_contiguity	PASS	contiguous
total_gap_count	PASS	108340298703
total_twin_count	PASS	5173760785
global_density	PASS	0.04775472143734025
cumulative_counts	PASS	exact
density_fields	PASS	match
summary_json_consistency	PASS	match

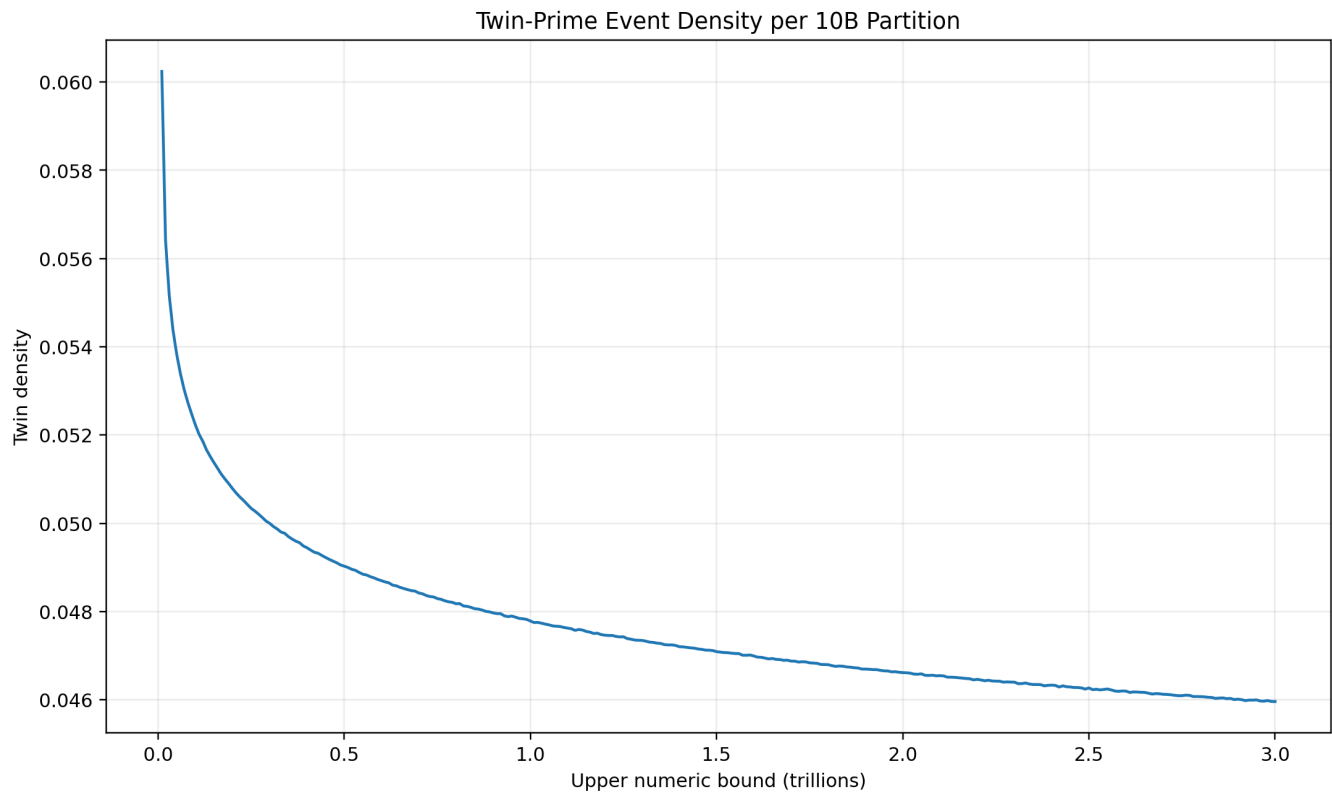
3. Scale summary

Upper bound	Partitions	Gaps	Twin events	Density
100,000,000,000	10	4,118,054,813	224,376,048	0.054485930418
500,000,000,000	50	19,308,136,142	986,222,314	0.051078069201
1,000,000,000,000	100	37,607,912,018	1,870,585,220	0.049739140506
2,000,000,000,000	200	73,301,896,139	3,552,770,943	0.048467654046
3,000,000,000,000	300	108,340,298,703	5,173,760,785	0.047754721437

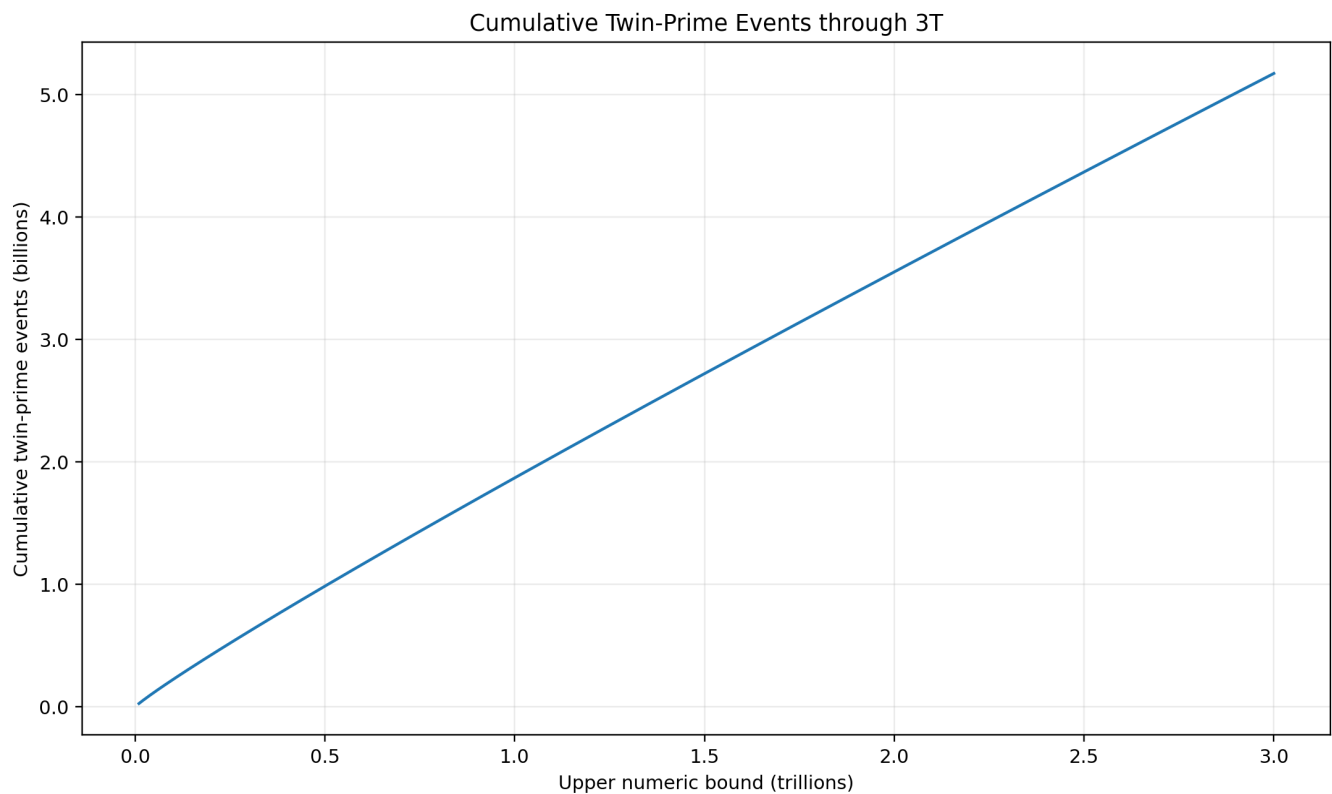
Twin-prime events per 10B partition



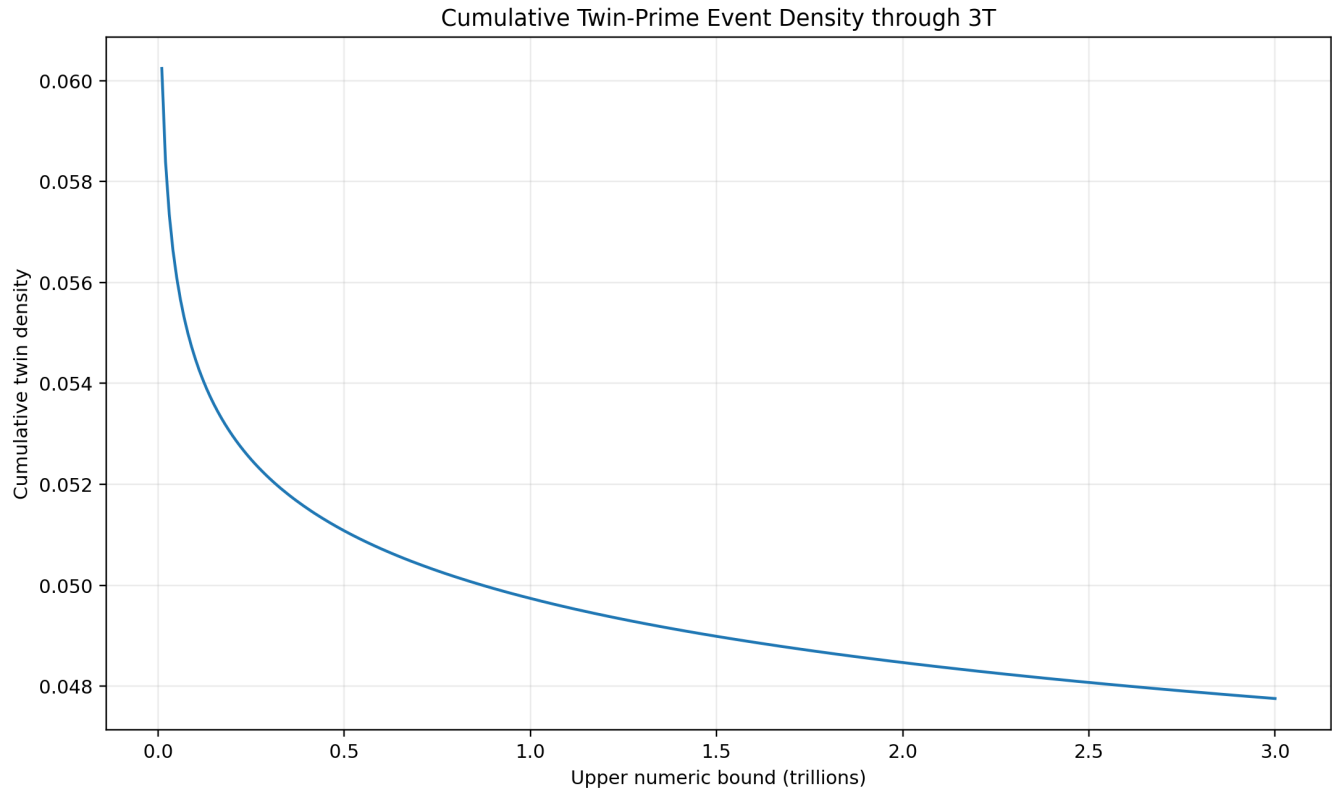
Twin-prime event density per 10B partition



Cumulative twin-prime events



Cumulative twin-prime event density



4. Interpretation and scope

The first 10B partition has local twin density 0.060240693848; the final 10B partition has local twin density 0.045962623212. The measured partition density declines gradually across the observed numeric scale, while cumulative twin-prime events grow monotonically. This report is a finite census over the accepted 1-3T repository. It does not constitute a proof of the Twin Prime Conjecture or an asymptotic theorem.

Twin-prime events are represented directly in the canonical PrimeNet coordinate system as $T = \{i : g(i) = 2\}$. The census therefore demonstrates how a scientifically important arithmetic family emerges naturally from the verified gap repository.